

THEJESH MALLIDI

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EXPERIENCE

Quantiphi Analytics

Irving, TX, USA

Machine Learning Engineer

06/2025 - Present

- Developed an **AI Governance** system that audits agentic use cases across legal, ethical, material, and security dimensions and monitors them in production for FinOps, ROI, and performance, standardizing a reusable agentic harness that cuts compliance review from days to minutes (~**200x** faster).
- Built an internal **Coding Agent CLI** (comparable to **Claude Code**) used org-wide, that analyzes codebases to flag security vulnerabilities, enforce internal engineering standards, auto-generate readme and documentation, and remediate bugs, cutting manual code-review effort by **80%**.
- Engineered an **LLM-powered document processing pipeline** for a Tier-1 telecom using self-hosted **LLaMA 3.1 70B** and **Gemini** models, transforming **2M+ contracts** from unstructured legalese into structured queryable **Knowledge graph**.
- Built a **GraphRAG-powered Document Assist chatbot** lets legal teams surface risks and key terms across 2M+ documents, cutting review cycles by **75%** with sub-second retrieval latency and **90%+** answer accuracy.
- Built a **unified agentic support system** using **Google ADK with Gemini Models** and **MCP servers**, serving **175+ Slack channels** through a single Cloud Run deployment that dynamically instantiates context-aware agents per request, reducing developer wait time for SME-driven issue resolution by over **90%**.
- Built an **agentic Slack workflow** across **175+ channels** (Google ADK, LiteLLM, MCP servers) that autonomously resolves internal employee issues without SME intervention, cutting wait times by **70%** and boosting productivity.

Michigan State University

East Lansing, MI, USA

Graduate Research Assistant - AI R&D

01/2024 – 05/2025

- Pretrained domain-adapted BERT (MLM)** with domain-aware masking (boosted mask probability for curated terminology) on 1M abstracts, delivering **+15% average accuracy** on downstream tasks.
- Built a **domain-specific knowledge graph** with over **2 million entities** and **1.4 million relationships** to accelerate hypothesis generation in plant science, leveraging **Named Entity Recognition (NER)** and **Relation Extraction**.
- Fine-tuned LLMs**, including **Qwen** and **LLaMA** variants using **LoRA** and created a **custom BERT-based multi-relation extraction model** for information extraction, achieving **94%** accuracy in entity extraction and **84%** in relation identification.
- Developed a **conversational research assistant** with **GraphRAG + GenAI workflows (LangGraph)** to reason over the KG and literature.
- Built large-scale topic analysis of **700k evolutionary biology research abstracts**, uncovering temporal trends, research shifts, and literature gaps to inform future studies; leveraged **BERT-based topic modeling** and **Agentic AI** to extract ~150 high-quality topics.

Quantiphi Analytics

Bangalore, India

Machine Learning Engineer

05/2021 - 07/2023

- Automated document processing pipelines** for Banking, Financial Services, and Insurance (BFSI) workflows using **computer vision** and **NLP-based models**, which reduced manual review time by **75%** and achieved **90%** data extraction accuracy.
- Delivered **10+ Proof of Concepts (PoCs)** on **document classification & extraction, object detection, image classification, and semantic segmentation** in collaboration with cross-functional teams of frontend developers and business stakeholders.
- Developed **insurance-focused predictive models** for risk assessment and claim outcome prediction using Logistic Regression, Decision Trees, Bagging, Random Forest, and Gradient Boosting Machines (GBM).
- Built and optimized real-time inference pipelines for Nvidia edge devices using **DeepStream** with **custom C++ plugins**, achieving **27 FPS** for latency-sensitive applications.
- Orchestrated the deployment of **20+ machine learning models** into production environments, leveraging **CI/CD pipelines** across **AWS SageMaker, GCP Vertex AI, and Kubernetes**, decreasing model deployment time by **30%**.

SKILLS

- Core Expertise:** Language Models (LLMs), Generative AI, Agentic AI, RAG, GraphRAG, Data Science, Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision, Large MultiModal AI
- Programming Languages:** Python, R, C++, C, Bash
- AI/ML Frameworks & Libraries:** Google ADK, LiteLLM, A2A protocol, Model Context Protocol (MCP), A2UI, AG-UI, LangGraph, LangChain, Hugging Face Transformers, PyTorch, TensorFlow, TensorRT, ONNX, LlamaIndex, Vector databases
- Big Data & Streaming:** Airflow, Hadoop Spark, Apache Kafka, Apache Beam
- Databases:** MySQL, PostgreSQL, Neo4j Graph DB, Elasticsearch, Redis
- Cloud & LLMOps:** GCP, AWS, Docker, Git, Flask, FastAPI, CI/CD Pipelines, Jenkins, LangFuse, Galileo, MLFlow, KubeFlow

EDUCATION

Michigan State University, USA

Master of Science in Data Science

08/2023 - 04/2025

GPA: 3.95/4.0